Type 1 Questions

Identification: Modulus sign appears at only one side e.g

$|x-4| > 3x+1$

Solution Method :

Eliminate Mod signs and introduce $\pm$ sign to split into two inequalities, illustrate the answer on the number line to obtain the final answer.

Note :

type 1 can also be solved by taking squares to both sides but this Method works better when inequality contains Mod signs at both sides, e.g $|x-4| > |3x+1|$, this is type 2.

TYPE 1 QUESTIONS

Q 1 Solve the inequality $|x - 2| < 3 - 2x$. 9709/03/M/J/03/Q3 [4]

ANS: $x < 1$.

Q2 Solve the inequality $|2^x - 8| < 5$. 9709/03/O/N/03/ Q1 [4]

Ans: $1.58 < x < 3.7$.

Q3 Solve the inequality $2x > |x - 1|$. 9709/03/M/J/06/Q2 [4]

$x < -1$ does not satisfy the inequality, $\therefore x > \frac{1}{3}$ **(Ans)**

Q4 Find the set of values of x satisfying the inequality $|3^x - 8| < 0.5$, giving 3 significant figures in your answer. 9709/03/O/N/06 / Q 1 [4]

$1.83 < x < 1.95$ **(Ans).**

Q5:

2 Solve the inequality $|x - 3| < 3x - 4$. [4]

$x > \frac{7}{4}$ **(Ans)**

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Q 6

1 Showing all necessary working, solve the equation $3|2^x - 1| = 2^x$, giving your answers correct to 3 significant figures. [4]

Ans : $x = 0.585$, -0.415.

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Q7

1 **(a)** Sketch the graph of $y = |x - 2|$. [1]

(b) Solve the inequality $|x - 2| < 3x - 4$. [3]

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answer $x > \frac{3}{2}$ only

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Q8

1 **(a)** Sketch the graph of $y = |4x - 2|$. [1]

(b) Solve the inequality $1 + 3x < |4x - 2|$. [4]

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answer $x < \frac{1}{7}$ [or] $x > 3$

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TYPE 2 QUESTIONS

Both sides have Mod sign e.g $|2x+3| > |x-1|$.

Method: square both sides, this results a quadratic ineq giving two possible answering. Squaring method might give an extra answer which might don't satisfy the original question, therefore, **MUST** check both the answers.

Q9 Solve the inequality $|2x + 1| < |x|$. 9709/03/M/J/04/Q2 [4]

Q 10 Given that a is a positive constant, solve the inequality 9709/03/O/N/05 /Q1

$$|x - 3a| > |x - a|.$$ [4]

Ans: $x < 2a$.

Q 11 Solve the inequality $|x - 2| > 3|2x + 1|$. 9709/03/M/J/08 /Q 1 [4]

ANS: $\therefore -1 < x < -\frac{1}{7}$.

Q 12 Solve the inequality $2|x-3| > |3x+1|$. 9709/32/O/N/10/Q1 [4]

$-7 < x < 1$ (Ans).

Q 13: Solve the inequality $|x| < |5 + 2x|$. 9709/32/M/J/11 / Q 1 [3]

$x < -5, \quad x > -\frac{5}{3}$ (Ans).

Q 14

Find the set of values of x satisfying the inequality $3|x-1| < |2x+1|$. 9709/32/O/N/12/Q1 [4]

$\frac{2}{5} < x < 4$ (Ans).

Q 15 Solve the equation $|x-2| = \left|\frac{1}{3}x\right|$. 9709/32/M/J/13/Q1 [3]

$x = 3$, or $x = \frac{3}{2}$ (Ans).

Question 16

$\frac{a}{4} < x < \frac{5a}{2}$ **(Ans).**

Find the set of values of x satisfying the inequality

$$|x + 2a| > 3|x - a|,$$

where a is a positive constant. **[4]**

[J14/P3/Q1]

Question 17

$-2 < x < \frac{1}{4}$ **(Ans).**

Solve the inequality $|2x-5| > 3|2x+1|$.

[4]

[N15/P3/Q1]

Q 18 Solve the inequality $3|2x - 1| > |x + 4|$. [4]

$x < -\frac{1}{7}, \quad x > \frac{7}{5}$ **(Ans).**

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Q 19 Solve the inequality $|2x - 1| < 3|x + 1|$. [4]

answer $x < -4,\ x > -\frac{2}{5}$

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Q 20 Solve the inequality $|3x - a| > 2|x + 2a|$, where a is a positive constant. [4]

answer $x > 5a$, $x < -\frac{3}{5}a$

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Q 21 Solve the inequality $|5x - 3| < 2|3x - 7|$. [4]

answer $x < \frac{17}{11}$, $x > 11$

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